	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Control Systems	3	0	0	3	3
	PC	Control Systems Lab	0	0	2	2	1
	PC	Linear Integrated Circuits and Applications	3	0	0	3	3
	PC	Linear Integrated Circuits and Applications Laboratory	0	0	2	2	1
	PC	Digital Communication	3	0	0	3	3
	PC	Digital Communication Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VI	OE	Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Computer and Communication Networks	3	0	0	3	3
	PC	Computer and Communication Networks Laboratory	0	0	2	2	1
	PC	Computer Organization and Microprocessor	3	0	0	3	3
	PC	Computer Organization and Microprocessor Lab	0	0	2	2	1
	PC	Microwave and Antenna	3	0	0	3	3
	PC	Microwave and Antenna Lab	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 4 (3-0-0)	3	0	0	3	3
	PE	Program Elective 5 (3-0-0)	3	0	0	3	3
	PC	Modern Wireless Communication	3	0	0	3	3
	PC	Modern Wireless Communication Lab	0	0	2	2	1
	PC	Embedded System Design	3	0	0	3	3
	PC	Embedded System Design Lab	0	0	2	2	1
	PC	Digital CMOS VLSI Design	3	0	0	3	3
	PC	Digital CMOS VLSI Design Lab	0	0	2	2	1
		Seminar					1
			18	**0**	**6**	**24**	**24**
Sem VIII		**Course Name**					
	Pro	Major/Comprehensive Project					**12**

Program Core Electives:

Semester	VLSI/Embedded systems	Communication Systems	Computer systems	Control
5	Digital Systems Design using HDL	Opto Electronics and Optical Communication		
6	Analog IC Design	Satellite Communication	Machine Learning and Applications	Introduction to Robotics
	Power Electronics	Information Theory and Coding	Database Management Systems	Modern Control Systems
7	Advanced Processors and SoCs	Advanced Communication Networks	Internet of Things	Drones: Design, Theory and Applications
	IC Technology	Radar and Navigation Systems	Deep Learning and Applications	
	Mixed Signal VLSI Design			

Semester 6

24ECE***T					Computer Communication and Networking					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To learn the fundamentals of networking layers and protocols.
- To be able to understand, analyze, and design computer communication systems.
- To develop various networking algorithms for enhancing the efficiency of existing networking protocols.
- To study real-life applications of different networking protocols.

UNIT I: PHYSICAL AND DATA LINK LAYERS	**12 Hrs.**
Introduction: Nuts and Bolts, Performance parameters: throughput, delay, etc., Layered Architecture (OSI and TCP/IP), Data link layer: Introduction, Media access protocols (ALOHA, CSMA based) , Ethernet 802.3, Token, ring 802.5, Reliability Issue: sliding window, Internetworking and Routing: Best effort Service, Virtual Circuits, IP Addressing, introduction to physical layer services, different types of modulation schemes, multiplexing and de-multiplexing techniques, parity, cyclic redundancy check.	
UNIT II: NETWORK LAYER	**12 Hrs.**
Internetworking and Routing: Routing Issues, Distance Vector and Link State routing, Intra and Inter Autonomous System Routing (OSPF, RIP, BGP), Broadcast and Multicast Routing Issues. Network layer data plane: Concept of data forwarding, Longest prefix matching, switching fabrics, packet scheduling, IP addressing: IPV4 and IPV6, concept of subnet, NAT.	
UNIT III: TRANSPORT LAYER AND APPLICATION LAYER	**10 Hrs.**
Transport Layer: End to end delivery issues, Reliable data transfers, Congestion Control, Traffic engineering and Quality of service, TCP, UDP, Socket Programming, Client-server applications, Echo and Chat applications, DHCP, HTTP, FTP, DNS, Peer to Peer file sharing application.	
UNIT IV: ADVANCED TOPICS	**8 Hrs.**
QoS over IP, IPV6, and Infrastructure-less networks: wireless ad hoc and sensor networks, and Internet of Things (IoT), Green computer networking, OFDM, 5G cellular network architecture, Flexible Optical Networks	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember concepts of protocol design at different network layers
CO2	:	Understand design and implementation of different networking protocols
CO3	:	Apply computer networking concepts to solve problems
CO4	:	Analyze computer networking protocols at different layers
CO5	:	Evaluate performance of different computer network protocols
CO6	:	Create computer networking applications

TEXT/REFERENCE BOOKS

1. James Kurose and Keith Rose, "Computer Networking: A Top Down Approach", Pearson Education
2. Larry L Peterson and Bruce S Davie, "Computer Networks: A Systems Approach", Elsevier
3. Andrew S Tanenbaum, "Computer Networks", Pearson Education
4. Behrouz A Forouzan, "Data Communication and Networking", McGraw Hill

24ECE***P					Computer Communication and Networking Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To learn the fundamentals of networking layers and protocols.
- To be able to analyze and design computer communication systems.
- To be able to implement various networking algorithms and analyze the performance using network simulators.

LIST OF EXPERIMENTS	
1	To study and prepare LAN cables (cross and straight), to configure LAN and perform Static Routing
2	Introduction to Socket Programming- Design and Implement client-server elements of a few network applications e.g. TCP/UDP client server, Echo client and server, Time client and server, Online Quiz and Buzzer Application, etc
3	Configure DHCP in a small LAN and understand its functionality using Wireshark/ Packet Tracer
4	Configure DNS in a small LAN and understand its functionality using Wireshark/ Packet Tracer
5	Understand functionality of HTTP using Wireshark/ Packet Tracer
6	Understand functionality of TCP and UDP using Wireshark/ Packet Tracer
7	Configure virtual LAN and understand its functionality using Wireshark/ Packet Tracer
8	Configure a wired network with Hosts, switches, hubs and routers using Packet Tracer
9	Configure a wireless LAN (Wi-Fi) using Packet Tracer.
10	Simulation of TCP/UDP connections and performance analysis
11	Simulation of a vehicular network using network simulator (NS3/NETSIM) and performance analysis
12	Simulation of 5G network using network simulator (NS3/NETSIM) and performance analysis

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember concepts of functionality of layered network architecture.
CO2	:	Understand various types of network topologies, network devices and their functions within a network.
CO3	:	Apply computer networking concepts to solve problems.
CO4	:	Analyze computer networking scenarios using network simulator tools.
CO5	:	Evaluate different computer network protocols practically.
CO6	:	Create computer networking applications and scenarios.

TEXT/REFERENCE BOOKS

1. James Kurose and Keith Rose, “Computer Networking: A Top Down Approach”, Pearson Education
2. Larry L Peterson and Bruce S Davie, “Computer Networks: A Systems Approach”, Elsevier
3. Andrew S Tanenbaum, “Computer Networks”, Pearson Education
4. Behrouz A Forouzan, “Data Communication and Networking”, McGraw Hil

24ICxxxT					Computer Organization and Microprocessor					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the design aspects of basic computer systems
- To explore the advanced features for throughput improvement
- To apply concepts in microprocessor system development
- To make a basis for operating systems by understanding memory management concepts.

UNIT 1: BASIC COMPUTER ORGANIZATION	**11 Hrs.**
Introduction to FSM, Moore, and Mealy machines. The basic architecture of micro-computer and Central processing unit, the concept of data, address and control bus, register transfer language, Stored program organization, instruction format, common bus system, instruction set completeness, timing, and control, instruction cycle, types of instructions, design of Basic central processing unit. Hardwired and microprogrammed control Units, Interrupts in basic computer.	
UNIT 2: ADVANCED FEATURES OF COMPUTER SYSTEM	**10 Hrs.**
Flynn's classification of computer systems, Parallel processing, pipelining, vector processing, arithmetic and instruction pipeline. Memory organization, cache and virtual memory management.	
UNIT 3: 8086 MICROPROCESSOR	**11 Hrs.**
Evolution of microprocessor, Pin diagram, signals, and register set of 8086. Addressing modes, instruction set, assembly language programming, Counters and Time delays, Stack, subroutine, Restart, Conditional Call and Return Instructions, interrupts of 8086.	
UNIT 4: 8 INTERFACING USING 8086	**10 Hrs.**
Address decoding techniques, interfacing concepts, IO mapped IO, Memory-mapped IO, interfacing of basic devices, interfacing PPI 8255, and Interrupt controller 8259.	

Total: 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Recall and describe the basic components of computer architecture, including the CPU, memory, and I/O devices, and understand their respective functions and interactions within a computer system.

CO2: Demonstrate the understanding of basic computer design, architecture and functionalities of the 8085 microprocessor, including its instruction set, addressing modes, and interfacing with peripherals.

CO3: Practice and debug assembly language programs for the 8085 microprocessor to perform specific tasks, demonstrating their ability to apply the knowledge of microprocessor architecture in practical programming scenarios.

CO4: Apprise the concept of programming, including data types, control structures, functions, and algorithms, and will be able to analyze and explain the functionality of different programming constructs

CO5: Evaluate the advanced architectural features and innovations in modern CPUs, such as pipelining, caching, multi-core processing, and parallel computing, and understand their impact on performance and efficiency.

CO6: Integrate and apply their knowledge of basic computer architecture design, 8085 microprocessor architecture, programming, and advanced CPU features to design, develop, and optimize a basic computer system or program, demonstrating their ability to create and innovate in the field of computer architecture and design.

TEXT/REFERENCE BOOKS

1. Morris M. Mano, "Computer System Architecture", Pearson Education, 3rd edition, 2017.
2. John P. Hayes, "Computer Architecture and Organization", McGraw Hill Education, 3rd edition, 1998.
3. William Stallings, "Computer Organization & Architecture", Pearson Education, 11th Edition, 2022.
4. Ramesh S. Gaonkar, "Microprocessor Architecture, Programming, and Applications with 8085", Penran International, 6th edition, 2013.

5. N. Senthil Kumar, M. Saravanan, S. Jeevanathan, S. K. Shah, “Microprocessors and Interfacing”, Oxford University Press, 2012.
6. Douglas Hall, “Microprocessor and Interfacing”, Mcgraw Hill Higher Education, 2005.

24ICxxxP					Computer Organization and Microprocessor Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES

- To understand design aspects of basic computer systems through simulation
- To apply concepts in microprocessor system development
- To do assembly language programming using 8086 instruction set to complete different tasks.

List of Experiments:

Following is the list of experiments, but it is not limited to:

1. To design a FSM with given sequence
2. To design and demonstrate the working of a massage on 7-segmnet display.
3. To design register with synchronous control inputs like clear, load and increment and with provision to handle different no of bit than common bus.
4. To design common bus systems using multiplexor
5. To design ALU with given specification
6. To design control unit for given CPU specification
7. To Write an assembly language code to implement data transfer instruction.
8. To write an assembly language code to store numbers in reverse order in memory location.
9. To write an assembly language code to implement arithmetic instruction.
10. To write an assembly language code to implement logical instruction set to carry out a task.
11. To write an assembly language code to implement the subroutine call.
12. To write an assembly language code to implement RIM and SIM to interface a peripheral.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Recall and describe the basic components of computer architecture, including the CPU, memory, and I/O devices, and understand their respective functions and interactions within a computer system.

CO2: Demonstrate the understanding of basic computer design, architecture and functionalities of the 8085 microprocessor, including its instruction set, addressing modes, and interfacing with peripherals.

CO3: Practice and debug assembly language programs for the 8085 microprocessor to perform specific tasks, demonstrating their ability to apply the knowledge of microprocessor architecture in practical programming scenarios.

CO4: Apprise the concept of programming, including data types, control structures, functions, and algorithms, and will be able to analyze and explain the functionality of different programming constructs

CO5: Evaluate the advanced architectural features and innovations in modern CPUs, such as pipelining, caching, multi-core processing, and parallel computing, and understand their impact on performance and efficiency.

CO6: Integrate and apply their knowledge of basic computer architecture design, 8085 microprocessor architecture, programming, and advanced CPU features to design, develop, and optimize a basic computer system or program, demonstrating their ability to create and innovate in the field of computer architecture and design.

TEXT/REFERENCE BOOKS

1. Morris M. Mano, “Computer System Architecture”, Pearson Education, 3rd edition, 2017.
2. John P. Hayes, “Computer Architecture and Organization”, McGraw Hill Education, 3rd edition, 1998.
3. William Stallings, “Computer Organization & Architecture”, Pearson Education, 11th Edition, 2022.
4. Ramesh S. Gaonkar, “Microprocessor Architecture, Programming, and Applications with 8085”, Penran International, 6th edition, 2013.
5. N. Senthil Kumar, M. Saravanan, S. Jeevanathan, S. K. Shah, “Microprocessors and Interfacing”, Oxford University Press, 2012.
6. Douglas Hall, “Microprocessor and Interfacing”, Mcgraw Hill Higher Education, 2005.

<Course Code>					Microwave and Antennas Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the concept of antenna radiation and fundamental parameters
- To analyze the wire, aperture, reflector, and microstrip antennas, and microwave devices.
- To design antennas and microwave components to meet given specifications.
- Test antennas and microwave components using the standard instruments/test benches.

UNIT I: INTRODUCTION TO ANTENNA AND MICROWAVE ENGINEERING	**8 Hrs.**
Microwave frequency bands, radiation mechanism, Near- and far-field regions, Fields and Power Radiated by an Antenna, Antenna Pattern Characteristics, Radiation mechanism, single wire, two wire, dipole, current distribution of thin wire antenna. Fundamental Parameters and Figures-of-Merit of Antennas, Friis Transmission Equation	
UNIT II: WIRE, APERTURE AND PRINTED ANTENNAS	**14 Hrs.**
Linear wire antennas: analysis, field equations, folded Dipole, Yagi-Uda antenna, image theory and monopole antenna. Aperture antennas: waveguide, horn and reflector antennas. Basic characteristics of microstrip antennas, feeding methods, methods of analysis, design of rectangular and circular patch antennas. Antenna Array: Two element arrays, N-element linear array, broadside array, ordinary end fire array, phased array.	
UNIT III: PASSIVE AND ACTIVE MICROWAVE DEVICES	**12 Hrs.**
Scattering parameters, waveguides, scattering parameters of wave-guide tees, directional couplers, circulators, and isolators. Microwave filters. Microwave Semiconductor Devices: Gunn Diodes, IMPATT diodes, Schottky Barrier diodes, PIN diodes, Microwave tubes: Klystron, TWT, Magnetron. Characteristics of Gunn Diode, Mode Characteristics of Klystron	
UNIT IV: MICROWAVE MEASUREMENTS	**8Hrs.**
Measuring Instruments: Principle of operation and application of VSWR meter, Power meter, Spectrum analyzer, Network analyzer, Measurement of Gain, Measurement of Impedance, Frequency, Power, VSWR, Q-factor, Dielectric constant, Scattering coefficients, Attenuation, S-parameters.	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember the basic fundamentals of electromagnetics, Maxwell‟s equations and Transmission Lines
CO2	:	Understand the concept of the basic antenna structures and microwave measurement
CO3	:	Apply the various figure of merits for antenna and microwave devices
CO4	:	Analyze the working of wire, aperture and microstrip antennas
CO5	:	Evaluate performance of the antenna structures and microwave components
CO6	:	Design antenna and microwave devices for modern wireless applications

TEXT/REFERENCE BOOKS:

1. C. A. Balanis, “Antenna Theory, Analysis and Design”, Wiley India Edition
2. D. Krauss, Antennas, McGraw Hill
3. Samuel Liao, Microwave Devices and Circuits, PHI
4. David M Pozar, Microwave Engineering, Wiley India Edition

24ICTxxxP					Microwave and Antenna Engineering Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	0	0	0	50	50	100

COURSE OBJECTIVES:

- To Impart electromagnetic and transmission line concepts
- To learn the radiating body behavior for wireless communication system
- To introduce the wave propagation mechanism in different modes
- To elaborate and familiarize the different modern RF technologies

List of Experiments:

Following is the list of experiments of experiments but it is not limited to it.

1. To measure position of fault in cable
2. To measure the frequency and wavelength using slotted line section and frequency meter.
3. To measure the Isolation and Insertion loss of Isolator and Circulator.
4. To study E-plane/H-plane/Magic Tee.
5. To measure Coupling Factor, Directivity and Isolation of directional coupler.
6. To measure VSWR and Reflection coefficient of different loads.
7. To study the characteristics of Klystron and Gunn diode.
8. To study the behavior of Transmission line: Waveguide/Coaxial cable/microstrip line.*
9. To study the S-parameter of directional coupler.*
10. To study the S-parameter of E-plane/H-plane Tee.*
11. To design and study MIC Based LPF.*

*** Some of these experiments need to be using simulation software.**

COURSE OUTCOMES:

On completion of the course, student will be able to

CO1: Remembering the fundamentals of electromagnetics, radiating structures, wag=ve propagation and modern RF techniques.

CO2: Understanding effect and outcome of RF waves propagating in different mediums including transmission line, free space and the under influence of RF Techniques.

CO3: Applying the concepts of transmission medium, radiating elements and wave propagation in developing new systems.

CO4: Analyzing the behaviour of wave and its impact on passive component properties in different transmission conditions

CO5: Evaluating different RF Technologies, components and mediums for RF Waves.

CO6: Design RF subsystems utilizing the radiating elements, passive components utilizing different mediums for developing different technologies.

TEXT/REFERENCE BOOKS

1. Samuel Y. Liao, “Microwave Devices and Circuits”, Pearson Education, 3rd edition, 2012.
2. C. A. Balanis, “Antenna theory: analysis and design”, John Wiley and sons, 3rd Edition, 2005.
3. M. L. Skolnik, “Introduction to Radar Systems”, Mc Graw Hill, 3rd edition, 2016.
4. K. D. Prasad, “Antenna and wave Propagation”, Satya Prakashan, 3rdedition, 2009.

Core Electives

Semester 6

24ECEXXXX					Machine Learning and Applications					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Introduce principles, algorithms, and applications of Machine Learning from mathematically motivated perspective.
- Work with different machine learning paradigms.
- Learn tools to apply machine learning algorithms to real data and performance evaluation.

UNIT 1 FUNDAMENTALS OF MACHINE LEARNING **11 Hrs.**

Concept of Data, Dimensionality, Concept of Features, Subset selection, Curse of dimensionality, PCA, Linear Discriminant
Analysis (LDA), various learning techniques, Statistical Decision Theory, Convex optimization, introduction

UNIT 2 FUNDAMENTALS REGRESSIONS AND CLASSIFICATION **11 Hrs.**

Neural Network, Back Propagation, Concept of Regression and classification, (One class classification, Multiclass Classification), KNN, , Convolutional Neural Network

UNIT 3 IMPROVING CLASSIFICATION **10 Hrs.**

Boosting, Bagging, Ensemble methods, Gradient boosting, Random Forests, HMM, Belief propagation

UNIT 4 CLUSTERING **10 Hrs.**

Clustering- Partitioning, Hierarchical, Density based, Gaussian Mixture Model, Expectation Maximization, K- Means clustering

Max. 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: List machine learning algorithm for the problem domain.
CO2: Compare different machine learning algorithms for the problems.
CO3: Carryout machine learning technique for classification and decision problems.
CO4: Analyse important parameters and structure of the machine learning algorithms.
CO5: Judge efficacy of machine learning algorithm.
CO6: Design and apply machine learning algorithm.

TEXT/REFERENCE BOOKS

1. Simon Haykin, Neural Networks and Learning Machines, Third Edition, Paper Back, Pearson India
2. Ethem Alpaydin, Introduction to Machine Learning, Third edition, PHI, 2015
3. Hastie, T., Tibshirani, R. and Friedman, J. The Elements of Statistical Learning. Springer. 2001
4. Karl Astrom, Richard Murray, „Feedback Systems: An Introduction to Scientists and Engineers*'*, Princeton University Press, 2008.
5. Kuo B.C., „Automatic Control Systems", PHI Pvt Ltd., New Delhi, 6th edition, 1991.
6. K. Hoffman and R. Kunze, „Linear Algebra", Prentice Hall, NJ, 1971.
7. P.R.Halmos, *'Finite Dimensional Vector Space'*, Springer, 1958.

<Course Code>					Information Theory and Coding					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To impart knowledge of information theory and source coding.
- To impart knowledge of channel coding methods.
- To impart knowledge of secure communication methods.

UNIT I: INFORMATION THEORY & SOURCE CODING	**12 Hrs.**
Concept & Measure of information, Entropy, Source Coding, Huffman and Shanon Fano coding, Uniqueness Property, Channel Models, Error Free Communication Over a noisy channel, Shannon‟s Theorem, Differential Entropy and Mutual information, Channel Capacity, Practical Communication System In Light of Shannon‟s Equation, Introduction to MIMO communication and Channel Capacity for MIMO Systems, Random Selection of Codes.	
UNIT II: ERROR CONTROL CODING (CHANNEL CODING)	**12 Hrs.**
Introduction, Linear block codes, Matrix Description, Syndrome Decoding, Error Probability, Hamming distance, Perfect codes and Hamming codes, Low Density Parity Check (LDPC) codes, Cyclic codes, Polynomial representation, Generation and Decoding of Cyclic codes, Burst Error Correcting and detecting code, Golay Codes, Cyclic Redundancy Check (CRC) Codes, Circuit Implementation of Cyclic Codes, Introduction to BCH codes, Reed-Solomon Codes.	
UNIT III: CONVOLUTION CODES & TRELLIS CODED MODULATION	**11 Hrs.**
Tree Codes and Trellis Codes, Polynomial Description, Generating Function, Matrix Descriptions, Viterbi Decoding Algorithms, Distance Bounds, Performance Bounds, Turbo Codes and Decoding, Trellis Coded Modulation, Decoder and Performance Evaluation, Space Time Trellis Codes.	
UNIT IV: CODING FOR SECURE COMMUNICATIONS	**07 Hrs.**
Introduction to Cryptography, An Overview of Encryption Techniques, Symmetric and Asymmetric Algorithms, Introduction to Advance Algorithms, Cryptanalysis.	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember basic concepts of information theory and coding.
CO2	:	Understand different types of coding techniques.
CO3	:	Apply various coding techniques for different applications.
CO4	:	Analyze encoding and decoding process of coding methods.
CO5	:	Evaluate and compare performance of various coding techniques.
CO6	:	Develop an error resilience, efficient and secure communication using coding schemes.

TEXT/REFERENCE BOOKS

1. Ranjan Bose, “Information Theory, Coding and Cryptography”, PHI.
2. B.P.Lathi, Zhi Ding “Modern Digital and Analog Communication Systems”, Oxford University Press.
3. Thomas M. Cover, Joy. A. Thomas, “Elements of Information Theory”, John Wiley.
4. Shu Lin and D.J. Costello Jr., “Error Control Coding”, Prentice Hall.

<Course Code>					Database Management Systems					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To learn the fundamentals of data models and to conceptualize and depict a database system using an ER diagram.
- To make a study of SQL and relational database design.
- To understand the internal storage structures which will help in physical DB design.
- To know the fundamental concepts of transaction processing- concurrency control techniques and recovery procedure.

UNIT I: CONCEPTUAL MODELING INTRODUCTION	**12 Hrs.**
Introduction - General introduction to database systems; Database - DBMS distinction, approaches to building a database, data models, database management system, three-schema architecture of a database, challenges in building a DBMS, various components of a DBMS.E/R Model - Conceptual data modeling - motivation, entities, entity types, various types of attributes, relationships, relationship types, E/R diagram notation, examples. Relational Data Model - Concept of relations, schema-instance distinction, keys, referential integrity, and foreign keys. Converting the database specification in E/R notation to the relational schema.	
UNIT II: RELATIONAL APPROACH and SQL QUERY - BASICS	**10 Hrs.**
Relational algebra operators: selection, projection, cross product, various types of joins, division, example queries, tuple relation calculus, domain relational calculus, Introduction to SQL, Data Definition of SQL, Basic structure of SQL queries, Basic SQL operations (rename, string operations, order by, where clause), Set operations, Null values, Aggregate functions, Nested Subqueries, Modification of Database, JOIN expressions, Views, Integrity constraints, Data types and Schemas,	
UNIT III: RELATIONAL DATABASE DESIGN	**10 Hrs.**
Dependencies and Normal forms, dependency theory - functional dependencies, Armstrong's axioms for FD's, closure of a set of FD's, minimal covers, definitions of 1NF, 2NF, 3NF and BCNF, decompositions and desirable properties of them, algorithms for 3NF and BCNF normalization, 4NF, and 5NF	
UNIT IV: TRANSACTION MANAGEMENT AND CONCURRENCY	**10 Hrs.**
Transaction processing: Transaction concept, transaction State, implementation of atomicity and durability, concurrent executions, serializability, recoverability. Concurrency Control: Lock-based protocols, timestamp-based protocols, validation-based protocols, multiple granularity, multiversion schemes, deadlock handling. Recovery: Failure classification, storage structure, recovery and atomicity, Log-Based recovery, shadow paging, recovery with concurrent transactions buffer management.	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand the data models, schemas, instances, view levels and database architecture for voluminous data storage
CO2	:	Remember the concept of Relational Algebra and Relational Calculus from set theory to represent queries
CO3	:	Apply the concepts of SQL queries for data aggregation, calculations, views, sub-queries, embedded queries manipulation
CO4	:	Illustrate the definition of Functional Dependencies, Inference rules, and minimal sets of FD"s to maintain data integrity.
CO5	:	Design the concepts of transaction, states, and ACID properties in data manipulation.
CO6	:	Design Build a database management system that satisfies relational theory and provides users with business queries.

TEXT/REFERENCE BOOKS

1. Silberschatz, Henry F. Korth, and S. Sudharshan, "Database System Concepts", 7th Ed, Tata McGraw Hill.
2. J. Date, A. Kannan and S. Swamynathan, "An Introduction to Database Systems", 8thed, Pearson Education
3. RamezElmasri and Shamkant B. Navathe, "Fundamentals of Database Systems", 7thEdition, Pearson/Addison-Wesley.

24ECExxxT					Introduction to Robotics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- Learn about diverse robotic systems, their types, key components, and significant applications.
- Grasp fundamental robot modeling principles and derive models using various methods.
- Understand concepts of control systems, controller design, operational principles in different robotics applications, and various estimation and motion planning techniques.
- Apply dynamic modeling and control to real robots in simulations and practical settings. Gain proficiency in robot programming and using open-source simulators for multi-robot applications.

UNIT I: ROBOTIC SYSTEMS AND TYPES	**08 Hrs.**
Introduction to robotics and autonomous systems, Common laws of robotics, Sensors and actuators, Perception and vision, Control systems and coordination approaches with real-world applications, Path and trajectory generation, Navigation and mapping. Types of robots: Stationary and Mobile robots, Manipulators and Industrial Robots, Unmanned Ground Vehicles (UGV), Wheeled mobile robots, Unmanned Aerial Vehicles (UAVs)/Drones: Fixed-wing & multirotor drones, Autonomous Underwater Vehicles (AUV), Bio- mimicking robots.	
UNIT II: ROBOT MODELLING	**10 Hrs.**
Basics: Configuration space, DOFs, Fully actuated, underactuated & overactuated robots; Kinematic modeling: Motions in 3D, Positions & rotations, Coordinate frames & transformations, Euler angles, Forward & inverse kinematics; Dynamical modeling: Kinetic & potential energy, Robot motions using Euler-Lagrange formulation; Modelling examples: 2-link robotic manipulators, Inverted pendulum on a cart, Slosh-container; Input-Output modeling: Transfer functions; State-space modeling: State variables, Linear & nonlinear system equations, Eigen structure.	
UNIT III: PLANNING AND CONTROL TECHNIQUES FOR ROBOTS	**13 Hrs.**
Controls: Disturbance effects, Uncertainties in models, Linear systems: Stability & time domain analysis, State-feedback control, Regulation & tracking, Nonlinear systems stability, Lyapunov based control design, Robust control designs, Optimization problems, Estimation techniques, Control in CPS & IoT systems, Using AI/ML for control systems. Motion Planning: Motivating examples, Basic ingredients of planning, Algorithms, Trajectory v/s path planning, Sensor-based motion planning, Motion planning via decomposition, Search algorithms using Graph theory, Shortest path & logic-based planning, Collision detection & avoidance.	
UNIT IV: EXPERIENCING THE ROBOTICS WORLD: SIMULATIONS, EXPERIMENTS &	**11 Hrs.**
Simulations: 2DOF underactuated robot, Quadrotor drone; Experimental robotics: How to build a robot using Arduino like microcontrollers, Practical aspects for ground & flying robots/drones. Introduction to Spark V and Firebird robots; Programming & simulators: Introduction to ROS: Concepts & basic programming, Introduction to Gazebo Simulator: The environment, Building a simple robot in Gazebo; Multiple robots based applications: Multi-agent systems, Surveillance, Monitoring, Package delivery, Agriculture, Smart solutions – Problem statements & challenges.	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to:

CO1	:	Remember and know about various real- world robotic systems & understand their types with their use.
CO2	:	Understand the system models using different modelling techniques for robotic systems .
CO3	:	Apply various control systems concepts for different robotics systems and in general.
CO4	:	Analyze the types of motion planning algorithms used in robotics.
CO5	:	Evaluate simulation based models, familiarize with practical robots and learn basic programming.
CO6	:	Create various cutting-edge single and multi-robot based application scenarios in new age

TEXT/REFERENCE BOOKS

1. Mark W. Spong, Seth Hutchinson, M. Vidyasagar, „Robot Modelling and Control", Wiley, 2006.
2. Roland Sigwart, Illah R. Nourbakhsh, „Introduction to Autonomous Mobile Robots", MIT Press, , 2004.
3. Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, Giuseppe Oriolo, 'Robotics: Modelling, Planning and Control", Springer, 2011.
4. Steven M. LaValle, „Planning algorithms", Cambridge University Press, 2006.

24ICxxxT					Satellite Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	0	0	100

COURSE OBJECTIVES

- To introduce the basics and benefits of satellite communication.
- To understand orbital mechanics principles of satellite communication systems and solve problems related to them.
- To understand the design of a satellite link and identify ways to improve the link performance.
- To study new technologies of satellite communication systems as per given specifications.
- To examine advanced technologies of satellite launching and describe the Indian satellite system.

UNIT-1: Introduction to Satellite Communication **12 Hrs.**

History, Overview of Satellite Communication, Types of Satellite, Types of Orbits, Satellite services, Advantages & Applications of Satellite communication, Satellite Life phases, Space Debris, Introduction to Geo-synchronous and Geo-stationary satellites. Orbital Mechanics, Kepler‟s Three Laws of Planetary Motion, Developing the Equations of the orbit, Look Angle Determination, Earth Stations, Orbital Perturbations, Orbital effects in Communication system performance.

UNIT-2: Optimal Design of Digital Receiver and Waveform Coding Techniques: **11 Hrs.**

Seven segments of Satellite communication, Attitude and Orbit control systems, Telemetry, Tracking and command control system, and Power supply system. Satellite Link Design: Basic transmission theory, System noise temperature and G/T ratio, Design of downlink and uplink, Design of satellite links for specified C/N.

UNIT-3: Design of Digital Communication Transceiver System **10 Hrs**.

VSAT, Direct broadcast satellite television, and radio, Satellite navigation and the Global positioning systems, GPS position location principle, GPS receivers and codes, Satellite Signal Acquisition, GPS navigation Message, GPS Signal Levels, Timing Accuracy, GPS Receiver Operation.

UNIT 4: Advanced Concepts and Way Forward: **9Hrs**.

Mechanism of Satellite launching, Launch Vehicles, Advanced launching techs like Space X, Intelligent Testing, Control and Decision making for Space, Inter Satellite Link. Indian Satellite Systems: History and Overview of Indian Satellite System, Achievements, GSLV, PSLV, Satellite in Cellular Communication.

Max. Hrs.: 42

COURSE OUTCOMES:

At the end of this course, students will be able to

CO1 - Remember thorough information on the conventional and upcoming satellite communication technology.

CO2 - Understand the fundamental aspects of satellite communications.

CO3 - Apply the knowledge of satellite communication systems to solve problems.

CO4 - Analyze the system performance of satellite communication systems.

CO5 - Evaluate the design of a satellite link and suggest enhancements to improve the link performance.

CO6 - Create satellite communication applications in the recent technological trends.

TEXT/REFERENCE BOOKS

1. T. Pratt, C. Bostian, and J. Allnutt, “Satellite Communications,” 2nd Ed., Wiley India, 2006. 2.
2. W. L. Pritchart, H. G. Suyderhoud, and R. A. Nelson, “Satellite Communication Systems Engineering,” 2nd Ed., Pearson Education, 2012.
3. G. Maral, M.Bousquet, and Z.Sun, “Satellite Communications Systems: systems, techniques and technology”, 5th edition, John Willy and sons
4. Tri T. Ha, “Digital Satellite Communications” McGraw-Hill, 2nd Edition
5. D. Roddy, “Satellite Communications,” 4th Ed., Tata McGraw-Hill Education, 2006.

23ECE305T					Modern Control Systems					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand various basic structures and tools of modern control system.
- To apply the knowledge state variable analysis in the analysis of control systems.
- To acquire the familiarity of some modern control techniques.
- To perform stability analysis and design capabilities of modern control systems for practical systems.

UNIT I: BASIC TOOLS AND STRUCTURES	08 Hrs.
Review of Feedback control Systems, Closed loop Transfer function and Disturbance rejection, Equilibrium and Stability notions, Degrees of Freedom and Configuration Space, Mathematical Structures: Rings, Groups, Vector Space, Subspace, Span, Linear dependence and independence, Basis, Dimension, Direct sum, Linear functional, Matrix differential Equations and State-space equations, Linear Systems Theory : Linearity, Time-invariance, Matrix exponential, System solutions, Eigen values and Eigen vectors, Similarity transformations, Jordan form, Controllability and Observability.	
UNIT II: DYNAMIC MODELLING AND BEHAVIORS OF MODERN CONTROL SYSTEMS	**10 Hrs.**
System engineering aspects, Cause and response in static and dynamic systems, Various modern control systems with examples, Dynamical equations of Modern control Systems, State-space models of various practical systems, Solving system equations, Energy based modelling of modern control systems, Nonlinear Systems, Qualitative analysis, Linearization, Local behaviour of nonlinear systems, Feedback Linearization.	
UNIT III: MODERN APPROACHES IN STABILITY ANALYSIS	**13 Hrs.**
Stability of linear approximations and phase portraits, Global and local stabilities, Lyapunov stability analysis, Stability theorems and corollaries, LaSalle‟s invariance principle, ISS stable, Analysis of real world modern control systems with examples.	
UNIT IV: MODERN CONTROLLERS AND ESTIMATORS	**11 Hrs.**
Review of PID control, State feedback control with pole placement, Output feedback control with pole placement, System uncertainties and its effects: Parametric, Functional, External disturbances, Need and types of robust controllers, Robust PID control, Robust control using Lyapunov methods, Sliding Mode Control, Basics of optimization methods for control design, LQR control, State estimation Problem: Kalman Filter, Observers, Luenberger observer, High-gain observer and Sliding mode observer. Future trends and the way forward - Control and estimation for modern applications like robotics, drones and Multi-agent systems, Data-driven control in AI applications.	
	TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to:

CO1	:	Identify various foundational concepts and tools for modern control systems.
CO2	:	Understand the system modelling approaches used in model based control designs.
CO3	:	Apply various tools for linearizations & tools for local analysis of nonlinear systems.
CO4	:	Analyze system stabilities for linearized systems & nonlinear systems using modern control tools.
CO5	:	Evaluate different types of modern controllers for robust & optimised control designs.
CO6	:	Create control designs using state-estimation algorithms.

TEXT/REFERENCE BOOKS

1. Chi-Tsong Chen, „Linear System Theory and Design‟, Oxford press, 1999.
2. Hassan K. Khalil, „Nonlinear Control*'*, Pearson, 2019.
3. Norman S. Nise, „Control Systems Engineering*'*, Wiley, 2015.
4. K. Ogata, „Modern Control Engineering*'*, PHI Pvt Ltd., New Delhi, 3rd Ed, 2000.

5. Karl Astrom, Richard Murray, „Feedback Systems: An Introduction to Scientists and Engineers*',* Princeton University Press, 2008.
6. Kuo B.C., „Automatic Control Systems", PHI Pvt Ltd., New Delhi, 6th edition, 1991.
7. K. Hoffman and R. Kunze, „Linear Algebra", Prentice Hall, NJ, 1971.
8. P.R.Halmos, *'Finite Dimensional Vector Space'*, Springer, 1958.

<Course Code>					Power Electronics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES –

- To introduce basic theory of power semiconductor devices and their application in power electronics circuits and systems
- To familiarize the working principle of AC-DC, DC-DC, DC-AC, AC-AC conversion circuits and their applications.
- To provide the basis for further study of power electronics circuits and systems.

UNIT I: POWER AMPLIFIERS AND POWER SEMICONDUCTOR DEVICES	**12 Hrs.**
Operation and analysis of power Amplifiers: Class A, B, C and push-pull configurations. Introduction to power electronics and its applications; overview of power electronic circuits and Systems Power semiconductor devices: Diodes, SCR, GTO and transistors (BJT, MOSFET and IGBT): Ratings, static and dynamic characteristics, drive and switching aid circuits; protection and cooling of power semiconductor devices	
UNIT II: AC-DC CONVERTERS	**11 Hrs.**
Principle of phase control AC-DC converter; single phase and three phase converter circuits with different types of loads; performance parameters analysis: power factor, ripple factor, form factor, harmonics, distortion and effect of source Inductance on performance; dual converters and their operation; power factor improvement techniques; applications of AC DC converter circuits.	
UNIT III: DC-DC AND AC-AC CONVERTERS	**12 Hrs.**
Introduction to DC-DC converters; working principle of step up and step-down chopper circuits; Design of chopper-based Buck converter, Boost converter, Buck-Boost and Cuk converter: circuit configuration and steady state time domain Analysis with different kinds of loads; design of inductors for DC-DC converters; AC-AC converter: single phase AC Voltage controllers, AC chopper; single phase cycloconverters; applications.	
UNIT IV: DC-AC CONVERTERS (INVERTERS):	**7 Hrs.**
Classification of inverters, single phase voltage source bridge inverter; performance parameter analysis with various loads; voltage control of single-phase inverters: PWM techniques; Three- phase bridge inverter: 180o and 120o conduction mode of operation with various loads; applications of inverter circuits.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Identify the power devices as per the usage of energy conversion and control.
CO2	:	Understand internal structure and operation of various power semiconductor devices.
CO3	:	Illustrate various converter topology/configuration with different types of loads.
CO4	:	Analyze the performance parameters for power electronic converters.
CO5	:	Select proper converter configurations for various power applications.
CO6	:	Develop a power converter topology for a specific industrial or domestic application.

TEXT/REFERENCE BOOKS

1. M. H. Rashid, “Power electronics: circuits, devices, and applications”, Pearson Education India, 2009
2. N. Mohan, T. M. Undeland, W.M. Robbins, “Power Electronics: Converters, Applications and Design”, Wiley India Edition, 2007.
3. M. D. Singh, K. B. Khanchandani, “Power Electronics”, 2nd Edn., Tata McGraw-Hill, 2007.
4. P.S. Bimbhra, “Power Electronics”, Khanna Publishers, New Delhi
5. P C Sen, “ Modern Power Electronics”, S Chand Publisher, 2013

<Course Code>					Analog IC Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	4	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand analysis and design of MOSFET circuits, current mirrors, and feedback.
- To understand analysis and design of CMOS operational amplifiers.
- To understand analysis and design of CMOS oscillators.

UNIT I: MOSFET-CIRCUITS CONFIGURATIONS AND CURRENT MIRRORS	**12 Hrs.**
Introduction to MOSFETs, MOSFET-Circuits Configurations: Common-Source, Source Follower, Common- Gate, Cascode, and Basic Differential Pair; Current Mirrors: Basic Current Mirrors, Cascode Current Mirrors, and Active Current Mirrors; Biasing Techniques: CS Biasing, CG Biasing, Source Follower Biasing, Differential Pair Biasing	
UNIT II: FEEDBACK	**10 Hrs.**
General Considerations: Properties of Feedback Circuits, Types of Amplifiers, Sense and Return Mechanisms, Feedback Topologies: Voltage-Voltage Feedback, Current-Voltage Feedback, Voltage-Current Feedback, Current-Current Feedback, Effect of Feedback on Noise, Effect of Loading in Feedback	
UNIT III: OPERATIONAL AMPLIFIERS	**10 Hrs.**
Performance Parameters, Two- Stage Operational Amplifier (Op Amp) Design, Gain Boosting Op Amp Design, Comparison, Stability and Frequency Compensation in Op Amps	
UNIT IV: OSCILLATORS	**10 Hrs.**
General Considerations, Ring Oscillators, LC Oscillators, Voltage-Controlled Oscillators, Concept of PLL.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Identify large-signal and small signal analysis of different MOSFET-based circuit configurations.
CO2	:	Understand the current mirrors.
CO3	:	Apply the CMOS feedback amplifier circuits.
CO4	:	Analyze the design of application-specific CMOS operational amplifiers.
CO5	:	Evaluate the design of robust and high-performance CMOS oscillators.
CO6	:	Create the design of advanced analog integrated circuits as per the application-specific requirements.

TEXT/REFERENCE BOOKS

1. Behzad razavi, “Design of analog CMOS integrated circuits,” 2nd edition, MC Graw hill, 2017.
2. J. Michael jacob, “Applications and design with analog integrated circuits,” phi, 2nd edition, 2004.
3. Paul r. Gray, paul j. Hurst, stephen h. Lewis, robert g. Meyer, “Analysis and design of analog integrated circuits,” 5th edition, wiley, 2009.
4. Sedra and smith, “Microelectronic Circuits,” oxford university press, 5th edition, 2005.